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# 1 SMART GLOVE To Predict Epilepsy



However, undetected epileptic attacks may lead to severe risks for the patients if unaccompanied or unattended on the spot. To help identify and even predict epileptic tendencies, Bengaluru-based TerraBlue XT has created a smart glove called TJay.

TJay smart glove can be used for the early detection and management of epilepsy. It can sense electrical signals from the human body and generate health-related data to keep track of one's health. It also provides information like health conditions during sleep, work and through daily chores. With all data, both in real time as well as historical, patients can get rid of uncertainties regarding their condition, while doctors can perform quicker and

more accurate diagnosis.

Several sensors attached to the glove pick up different data including subtle changes to the sympathetic and parasympathetic nervous system through the palm. These read data like skin conductance, muscle tone, heart rate, oxygen saturation, temperature and motion. Data is then transferred to the cloud through a personalised data gateway, which comes with TJay.

The gateway and smartphone can be synced with TJay using Wi-Fi and Bluetooth.

TerraBlue team has also developed a proprietary algorithm using machine learning and artificial intelligence (AI) to process and analyse the data, derive necessary insights and develop predictions for a probable epileptic onset. This enables predictive alerts and enables preventive management of the situation.

Patients can be given emergency medication to ease the problem or taken to a medical facility without delay. Nominated emergency contacts are notified immediately with location details of the patients when a seizure is about to start. Analytics generated are used by doctors to customise management plans for patients and to take informed remedial decisions. Moreover, this enables doctors to remotely monitor and consult patients, and even provide doorstep medical attention.

All data and reports can be accessed using TJay mobile application or over Web. Currently, TJay is best suited for detecting generalised tonic clonic seizures, complex partial seizures and partial seizures. TerraBlue has applied for a patent for TJay. TJay is right now up for preorder on the company website.

The innovators also have a community platform called TJay Circle of Care, which brings patients, caregivers, doctors and medical institutions to a common platform to discuss and get suggestions for problems related to epilepsy.

—Paromik Chakraborty, technical journalist, EFY

**E**pilepsy is a common, yet serious, neurological disorder that leads to associated health problems and risks. It is found in more than 50 million people worldwide—more than 2.4 million people are diagnosed with it every year. It is recognised most commonly through recurrent seizures and, at times, loss of bowel control and loss of consciousness. Epilepsy is treatable if diagnosed properly.